QUESTION PAPER - SET 5

SECTION 1 - PERSONAL DATA

1. Name

____

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2. Mobile Number

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3. Class 10 Overall Percentage

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4. Class 12 PCM Percentage

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5. What is your percentage in Pre Sea Training till 4th Semester?

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6. Tell us about your strengths.

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7. Tell us about your weaknesses.

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8. Have you participated in any extracurricular activities during your Pre Sea Training? If

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9. yes, share details.

10. Tell us about your family.

11. Where do you see yourself after 5 years?

SECTION 2 - COMPREHENSION PASSAGE

Archaeology as a profession faces two major problems.

First, it is the poorest of the poor. Only paltry sums are available for excavating and even less is available for publishing the results and preserving the sites once excavated. Yet archaeologists deal with priceless objects every day.

Second, there is the problem of illegal excavation, resulting in museum-quality pieces being sold to the highest bidder.

I would like to make an outrageous suggestion that would at one stroke provide funds for archaeology and reduce the amount of illegal digging. I would propose that scientific archeological expeditions and governmental authorities sell excavated artifacts on the open market. Such sales would provide substantial funds for the excavation and preservation of archaeological sites and the publication of results. At the same time, they would break the illegal excavator's grip on the market, thereby decreasing the inducement to engage in illegal activities.

You might object that professionals excavate to acquire knowledge, not money. Moreover, ancient artifacts are part of our global cultural heritage, which should be available for all to appreciate, not sold to the highest bidder. I agree. Sell nothing that has unique artistic merit or scientific value. But, you might reply, everything that comes out of the ground has scientific value. Here we part company. Theoretically, you may be correct in claiming that every artifact has potential scientific value. Practically, you are wrong.

I refer to the thousands of pottery vessels and ancient lamps that are essentially duplicates of one another. In one small excavation in Cyprus, archaeologists recently uncovered 2,000 virtually indistinguishable small jugs in a single courtyard, even precious royal seal impressions known as melekh handles have been found in abundance — more than 4,000 examples so far.

The basement of museums is simply not large enough to store the artifacts that are likely to be discovered in the future. There is not enough money even to catalogue the finds; as a result, they cannot be found again and become as inaccessible as if they had never been discovered. Indeed, with the help of a computer, sold artifacts could be more accessible than are the pieces stored in bulging museum basements. Prior to sale, each could be photographed and the list of the purchasers could be maintained on the computer A purchaser could even be required to agree to return the piece if it should become needed for scientific purposes. It would be unrealistic to suggest that illegal digging would stop if artifacts were sold in the open market. But the demand for the clandestine product would be substantially reduced. Who would want an unmarked pot when another was available whose provenance was known, and that was dated stratigraphically by the professional archaeologist who excavated it?

1. The primary purpose of the passage is to propose

A. an alternative to museum display of artifacts
B. a way to curb illegal digging while benefiting the archaeological profession
C. a way to distinguish artifacts with scientific value from those that have no such value
D. the governmental regulation of archaeological sites E.a new system for cataloging duplicate artifacts

2. The author implies that all of the following statements about duplicate artifacts are true EXCEPT:

A. A market for such artifacts already exists.
B. Such artifacts seldom have scientific value.
C. There is likely to be a continuing supply of such artifacts.
D. Museums are well supplied with examples of such artifacts. E. Such artifacts frequently exceed in quality in comparison to those already cataloged in museum collections

3. Which of the following is mentioned in the passage as a disadvantage of storing artifacts in museum basements?

A. Museum officials rarely allow scholars access to such artifacts.
B. Space that could be better used for display is taken up for storage.
C. Artifacts discovered in one excavation often become separated from each other. D Such artifacts are often damaged by variations in temperature and humidity. E. Such artifacts' often remain uncatalogued and thus cannot be located once they are put in storage

4. The author's argument concerning the effect of the official sale of duplicate artifacts on illegal excavation is based on which of the following assumptions?

A. Prospective purchasers would prefer to buy authenticated artifacts.
B. The price of illegally excavated artifacts would rise.
C. Computers could be used to trace sold artifacts.
D. Illegal excavators would be forced to sell only duplicate artifacts. E. Money gained from selling authenticated artifacts could be used to investigate and prosecute illegal excavators

5. The author anticipates which of the following initial objections to the adoption of his proposal?

A. Museum officials will become unwilling to store artifacts.
B. An oversupply of salable artifacts will result and the demand for them will fall.
C. Artifacts that would have been displayed in public places will be sold to private collectors.
D. Illegal excavators will have an even larger supply of artifacts for resale. E. Counterfeiting of artifacts will become more commonplace

SECTION 3 - COMPREHENSION PASSAGE

The Sundarbans, located in the delta region of India and Bangladesh, is the largest mangrove forest in the world. It is home to diverse wildlife, including the famous Royal Bengal Tiger. The mangroves act as a natural barrier, protecting coastal areas from cyclones and storm surges. They also provide livelihoods for local communities through fishing, honey collection, and eco-tourism. However, climate change poses a serious threat to the Sundarbans. Rising sea levels, increasing salinity, and frequent storms are causing habitat loss.

Conservation projects are being implemented, focusing on planting more mangroves, reducing illegal logging, and promoting sustainable fishing practices. International cooperation is considered vital for the long-term preservation of this unique ecosystem.

1. Which of the following is NOT listed as a livelihood activity for local communities in the Sundarbans, according to the passage? (1 Point) Select the option that is not mentioned as a source of livelihood.

A. Agriculture
B. Honey collection
C. Fishing
D. Eco-tourism

2. Which of the following is a direct impact of climate change on the Sundarbans ecosystem? (1 Point) Select the option that best represents a climate change effect described in the passage.

A. Increasing salinity of water
B. Introduction of new wildlife species
C. Growth in honey collection
D. Expansion of fishing zones

3. What is one of the key protective roles played by the mangrove forests in the Sundarbans? (1 Point) Select the option that best describes how mangroves benefit the coastal environment.

A. They increase the frequency of tidal waves.
B. They are a major source of underground minerals.
C. They supply drinking water to nearby villages.
D. They act as a shield against cyclones and storm surges.

4. What is the primary objective of the conservation efforts described in the Sundarbans passage? (1 Point) Select the option that best reflects the main goal of these projects.

A. To increase honey production for export
B. To encourage local communities to move to urban areas
C. To restore and safeguard the mangrove forests and their biodiversity
D. To expand commercial fishing operations

5. Why does the passage emphasize the necessity of international cooperation for preserving the Sundarbans? (1 Point) Select the option that best explains the importance of cross-border collaboration.

A. The Sundarbans ecosystem spans across India and Bangladesh, requiring joint efforts.
B. Local communities are unwilling to support conservation projects.
C. The region attracts international tourists who fund conservation.
D. Climate change is only a concern for a few developed countries.

SECTION 4 - ENGLISH APTITUDE

1. What is Software Testing?

a) Analyzing documentation

b) Writing code

c) Executing a system to identify bugs and errors

d) Compiling source code

2. What is the purpose of software testing in the SDLC?

a) To improve the user interface

b) To speed up the release process

c) To reduce the development time

d) To detect defects and ensure the software meets requirements

3. What is a test case?

a) A list of all the bugs in the software

b) A document that describes how a test should be executed

c) A report of testing results

d) The final version of the software

SECTION 5 - NUMERICAL APTITUDE

1. What is Functional Testing?

a) Checking database constraints

b) Validating that features work as expected according to requirements

c) Analyzing code logic

d) Testing the system’s performance

2. What is User Acceptance Testing (UAT)?

a) Debugging system errors

b) Running automated test execution

c) Validating business requirements with end-users

d) Testing system architecture

3. What is the role of a test manager?

a) To write code for the software

b) To deploy the software to production

c) To develop the test environment

d) To oversee the testing process and ensure test cases are executed

SECTION 6 - MARITIME KNOWLEDGE

1. What does the term “lee shore” refer to in seamanship?

A. A shore protected from wind

B. A shore facing the wind

C. A dangerous shore onto which wind blows

D. A shallow coastal area

2. Which frontal type is MOST commonly associated with the process shown in the diagram?

A. Stationary front

B. Warm front

C. Cold front

D. Occluded front

3. At what temperature does the air reach saturation, causing water vapor to begin condensing into liquid droplets?

A. Temperature at which air becomes saturated

B. Temperature of condensation clouds

C. Lowest night temperature

D. Temperature at which rain begins

4. Which emergency equipment is used to send distress signals to nearby ships?

A. EPIRB

B. Gyro Compass

C. Echo Sounder

D. Barograph

5. In navigation, what does “dead reckoning” primarily rely upon?

A. Radar observations

B. Celestial sightings

C. Estimated course and speed

D. Satellite communication

6. Which factor mainly affects ocean tides?

A. Wind speed

B. Earthquake activity

C. Gravitational pull of the Moon and Sun

D. Atmospheric pressure only

7. Which of the following corrections must be applied to a sextant reading to accurately determine the observed altitude (Ho) of a celestial body?

A. Index error, dip, refraction, semi-diameter & parallax

B. Index error and dip only

C. Refraction only

D. Dip and refraction only

8. What is the full form of GPS?

A. Global Positioning System

B. General Port System

C. Global Port Signal

D. Geographic Position Source

9. What is the primary purpose of a ship’s freeboard?

A. To increase cargo capacity

B. To improve engine efficiency

C. To provide reserve buoyancy and safety

D. To reduce fuel consumption

10. What is the standard distress frequency used in marine VHF communication?

A. Channel 6

B. Channel 13

C. Channel 16

D. Channel 21

11. A vessel departs Felixstowe, England at 1000 hrs local time (GMT+0) on 2nd March 2014, bound for Kolkata, India, a distance of 10,080 nautical miles. If the vessel maintains a constant speed of 10 knots, what will be the vessel's estimated time of arrival (ETA) at Kolkata in local time (GMT+ 5.5)?

A. 13th April 2014, 1000 hrs LT

B. 13th April 2014, 0430 hrs LT

C. 13th April 2014, 1530 hrs LT

D. 13th April 2014, 1200 hrs LT

12. In the Northern Hemisphere, what is the typical direction of wind circulation around a low-pressure area?

A. Anticlockwise and inward

B. Clockwise and inward

C. Clockwise and outward

D. Anticlockwise and outward

13. In ship stability, GM refers to:

A. Gross Measurement

B. General Metric

C. Metacentric Height

D. Gravity Measurement

14. What does “starboard” refer to on a ship?

A. Left side

B. Front side

C. Right side

D. Rear side

15. A vessel alters course from 090° to 180°. What is the amount of alteration?

A. 45°
B. 90°
C. 180°
D. 270°

16. In celestial navigation, what is the First Point of Aries ( ) primarily used to measure?

A. Measuring True Altitude
B. Measuring Azimuth
C. Measuring Right Ascension
D. Measuring Longitude

17. According to ship safety regulations, which of the following must be made weathertight as part of the Conditions of Assignment?

A. Hatchways situated on the freeboard deck
B. Sounding pipes on deck
C. Cargo hold ventilators below deck
D. Scuppers above deck

18. Which instrument is used to measure the salinity of seawater?

A. Hygrometer
B. Sextant
C. Salinometer
D. Barograph

19. Which chart projection is most commonly used for marine navigation?

A. Gnomonic Projection
B. Mercator Projection
C. Polar Projection
D. Cylindrical Equal Area Projection

20. What is the purpose of a ship's compass?

A. To measure sea depth
B. To determine direction
C. To measure wind speed
D. To detect underwater objects

21. What is the main function of a life raft?

A. Cargo storage
B. Emergency accommodation after abandoning ship
C. Firefighting
D. Navigation support

22. What does "port side" mean on a vessel?

A. Right side
B. Left side
C. Front side
D. Upper deck

23. At which location on Earth does the Coriolis effect reach its greatest strength?

A. At the Equator
B. At the Tropics
C. At 30° latitude
D. At the Poles

24. What is the primary navigational purpose of observing the Sun at its meridian passage?

A. Applying time correction
B. Finding longitude
C. Calculating compass error
D. Determining latitude

25. Which navigation light is carried by a vessel not under command?

A. Two all-round red lights in a vertical line
B. One all-round white light
C. Green and red sidelights only
D. Two all-round green lights

26. The observer's prime vertical passes through the east and west points of which of the following?

A. Rational horizon
B. Equinoctial
C. Celestial meridian
D. Great circle

27. What is the full form of CSK?

A. Chennai Super Kings
B. DDDD
C. SSSS
D. FFFF

28. Which type of weather front is most commonly linked to severe thunderstorms, heavy rain, and sudden gusty winds?

A. Cold front
B. Occluded front
C. Warm front
D. Stationary front

29. Which instrument is used to determine the depth of water beneath a vessel?

A. Barometer
B. Echo Sounder
C. Sextant
D. Hygrometer

30. Which method is most commonly used to determine compass error during sunrise or sunset?

A. Latitude by meridian passage

B. Azimuth

C. Sextant altitude

D. Amplitude